



Chart.js

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Chart.js is a free JavaScript library for making HTML-based charts. It is one of the simplest visualization libraries for JavaScript, and comes with the following built-in chart types:

- Scatter Plot
- Line Chart
- Bar Chart
- Pie Chart
- Donut Chart
- Bubble Chart
- Area Chart
- Radar Chart
- Mixed Chart

How to Use Chart.js?

Chart.js is easy to use.

First, add a link to the providing CDN (Content Delivery Network):



Then, add a `<canvas>` to where you want to draw the chart:

```
<canvas id="myChart" style="width:100%;max-width:700px"></canvas>
```

The canvas element must have a unique id.

That's all!

Typical Scatter Chart Syntax:

```
const myChart = new Chart("myChart", {  
  type: "scatter",  
  data: {},  
  options: {}  
});
```

Typical Line Chart Syntax:

```
const myChart = new Chart("myChart", {  
  type: "line",  
  data: {},  
  options: {}  
});
```

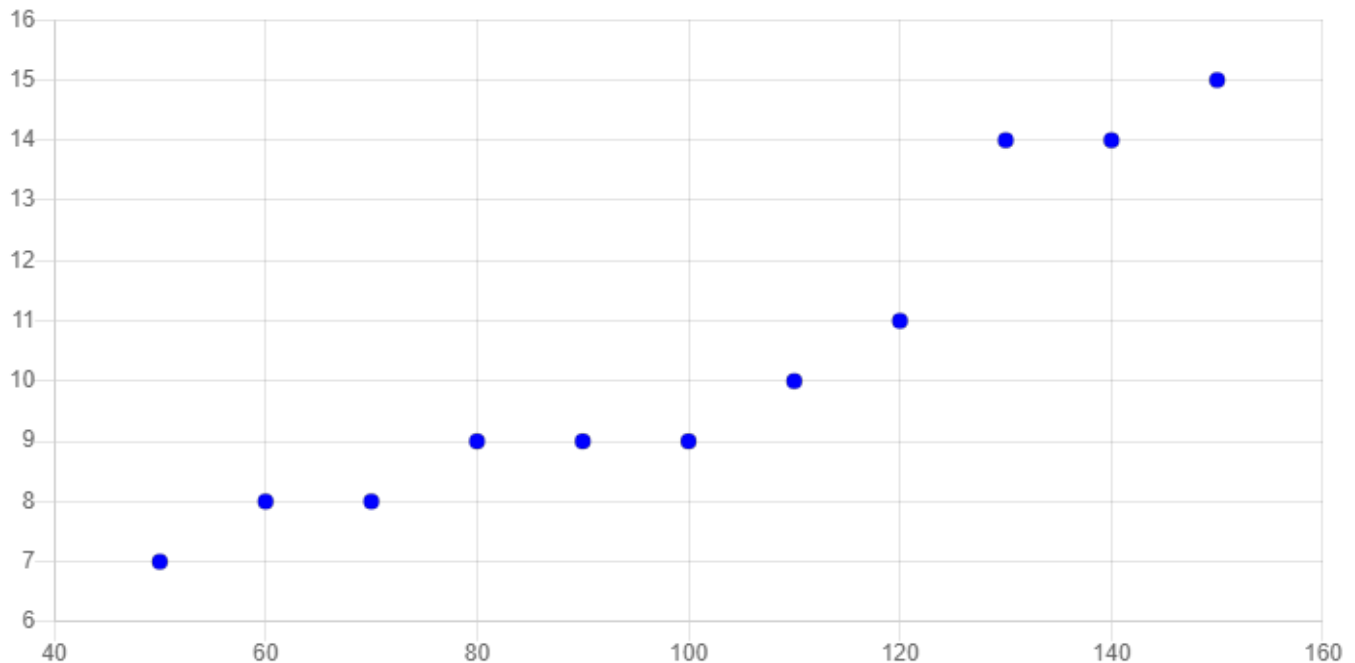
Typical Bar Chart Syntax:

```
const myChart = new Chart("myChart", {  
  type: "bar",  
  data: {},
```



Scatter Plots

House Prices vs. Size



Source Code

```
const xyValues = [  
  {x:50, y:7},  
  {x:60, y:8},  
  {x:70, y:8},  
  {x:80, y:9},  
  {x:90, y:9},  
  {x:100, y:9},  
  {x:110, y:10},  
  {x:120, y:11},  
  {x:130, y:14},  
  {x:140, y:14},  
  {x:150, y:15}  
];
```



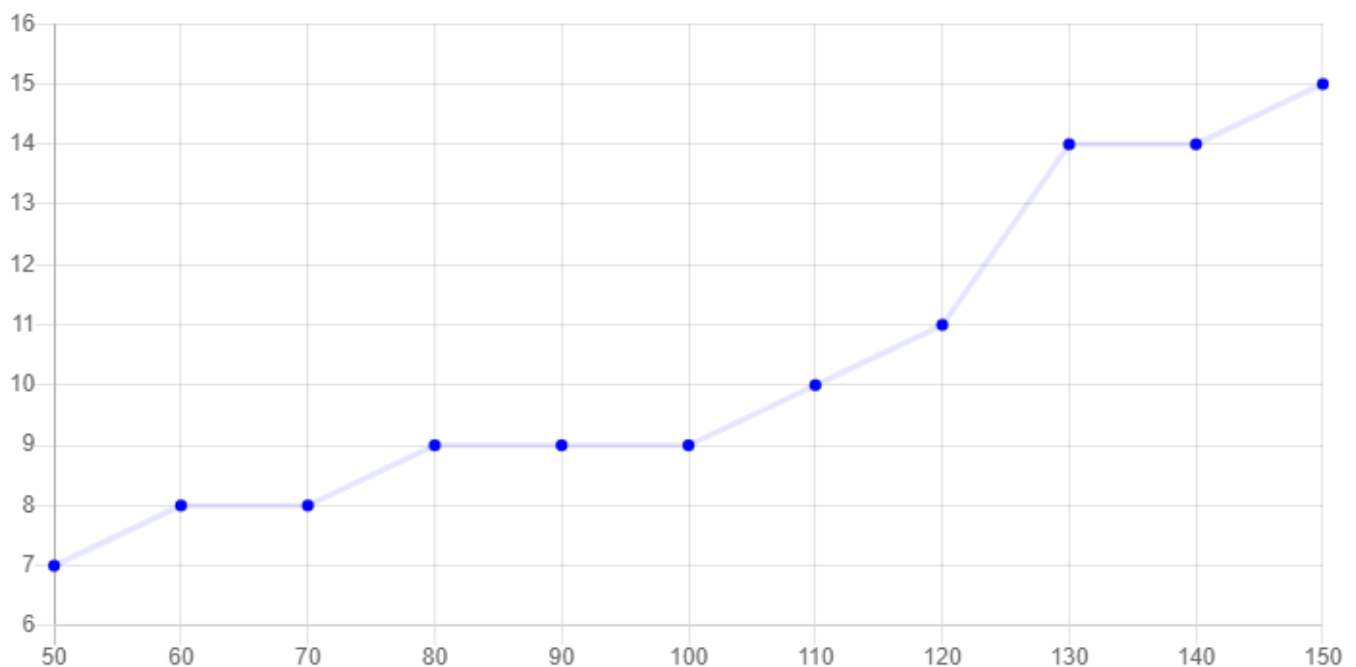
```
    datasets: [{  
      pointRadius: 4,  
      pointBackgroundColor: "rgba(0,0,255,1)",  
      data: xyValues  
    }]  
  },  
  options:{...}  
});
```

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Line Graphs

House Prices vs. Size





```
const xValues = [50,60,70,80,90,100,110,120,130,140,150],
const yValues = [7,8,8,9,9,9,10,11,14,14,15];

new Chart("myChart", {
  type: "line",
  data: {
    labels: xValues,
    datasets: [{
      backgroundColor: "rgba(0,0,255,1.0)",
      borderColor: "rgba(0,0,255,0.1)",
      data: yValues
    }]
  },
  options: {...}
});
```

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If you set the borderColor to zero, you can **scatter plot** the line graph:

```
borderColor: "rgba(0,0,0,0)",
```

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Multiple Lines



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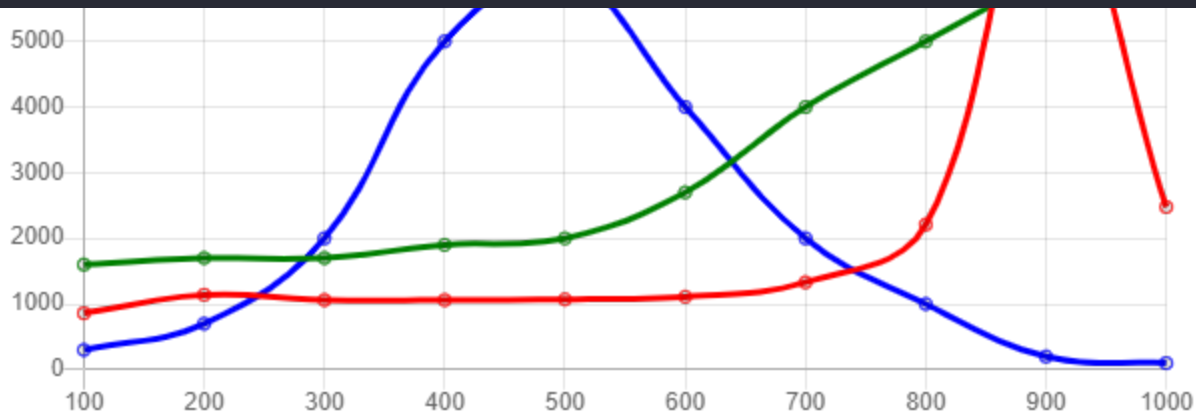
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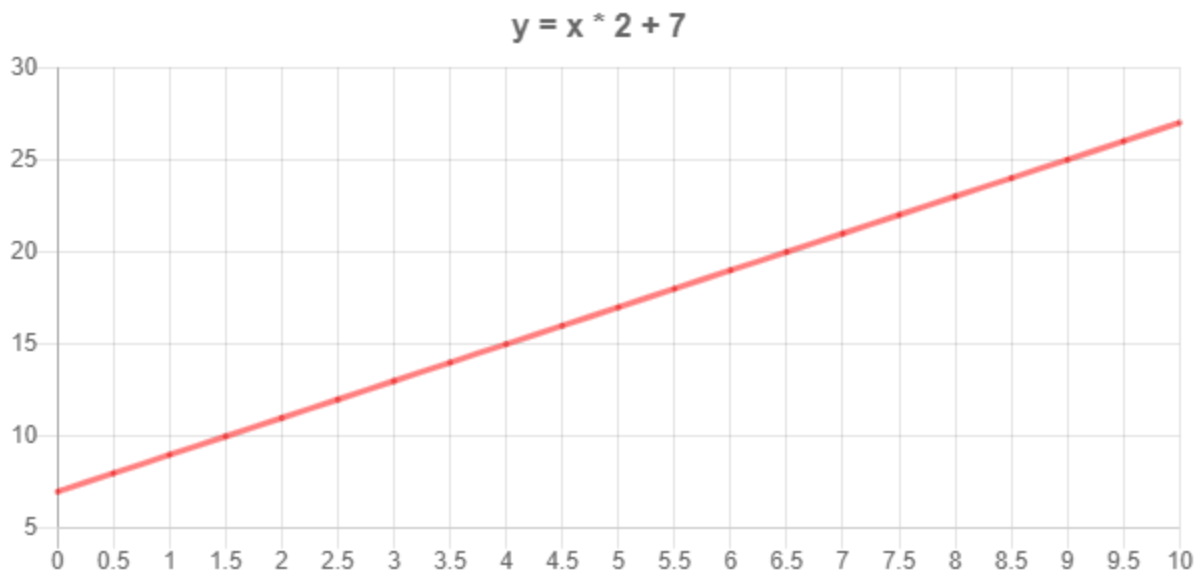
Source Code

```
const xValues = [100,200,300,400,500,600,700,800,900,1000];

new Chart("myChart", {
  type: "line",
  data: {
    labels: xValues,
    datasets: [{
      data: [860,1140,1060,1060,1070,1110,1330,2210,7830,2478],
      borderColor: "red",
      fill: false
    },{
      data: [1600,1700,1700,1900,2000,2700,4000,5000,6000,7000],
      borderColor: "green",
      fill: false
    },{
      data: [300,700,2000,5000,6000,4000,2000,1000,200,100],
      borderColor: "blue",
      fill: false
    }]
  },
  options: {
    legend: {display: false}
  }
});
```



Linear Graphs



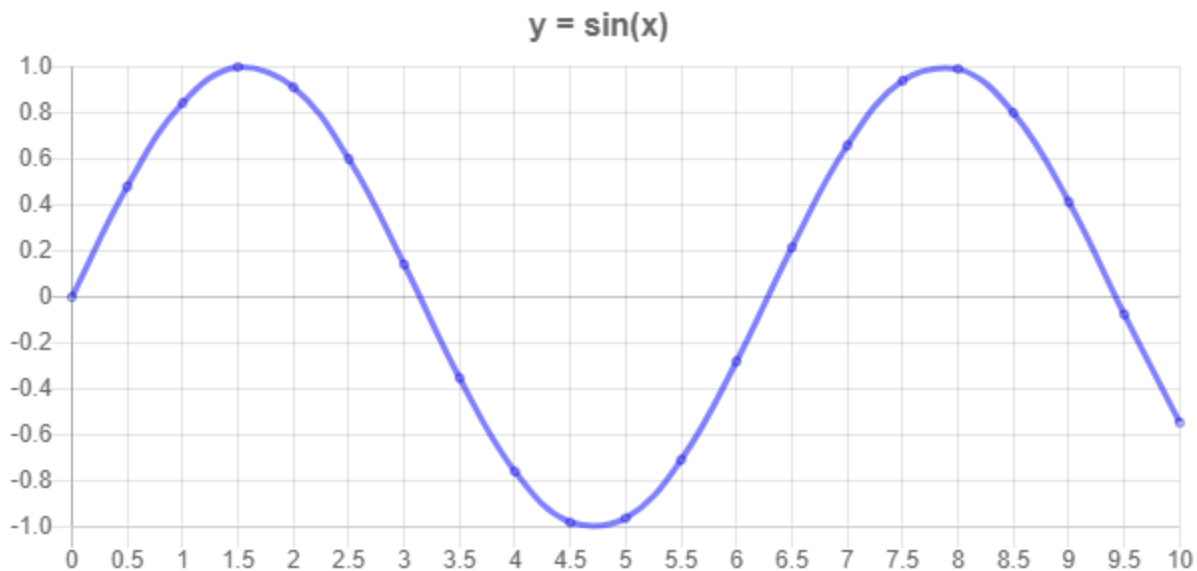
Source Code

```
const xValues = [];  
const yValues = [];  
generateData("x * 2 + 7", 0, 10, 0.5);  
  
new Chart("myChart", {  
  type: "line",  
  data: {  
    labels: xValues,  
    datasets: [{  
      fill: false,  
      pointRadius: 1,  
      borderColor: "rgba(255,0,0,0.5)",  
      data: yValues  
    }]  
  },  
  options: {...}  
});
```

```
}  
}
```

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Function Graphs

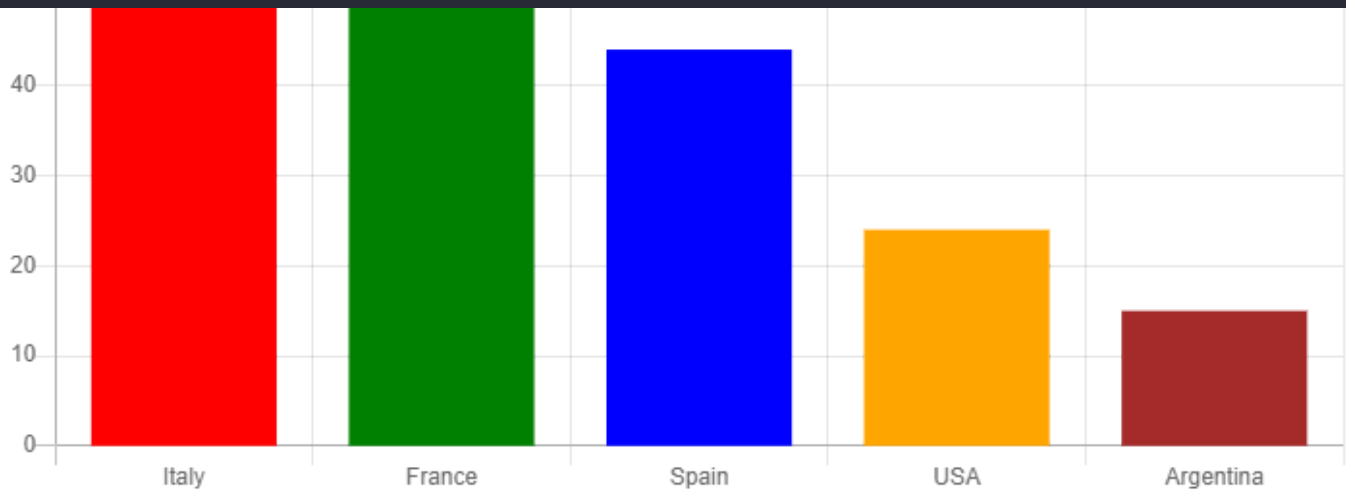


Same as Linear Graph. Just change the generateData parameter(s):

```
generateData("Math.sin(x)", 0, 10, 0.5);
```

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Bar Charts



Source Code

```
var xValues = ["Italy", "France", "Spain", "USA", "Argentina"];
var yValues = [55, 49, 44, 24, 15];
var barColors = ["red", "green", "blue", "orange", "brown"];

new Chart("myChart", {
  type: "bar",
  data: {
    labels: xValues,
    datasets: [{
      backgroundColor: barColors,
      data: yValues
    }]
  },
  options: {...}
});
```

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Color only one bar:



Same color all bars:

```
var barColors = "red";
```

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Color Shades:

```
var barColors = [  
  "rgba(0,0,255,1.0)",  
  "rgba(0,0,255,0.8)",  
  "rgba(0,0,255,0.6)",  
  "rgba(0,0,255,0.4)",  
  "rgba(0,0,255,0.2)",  
];
```

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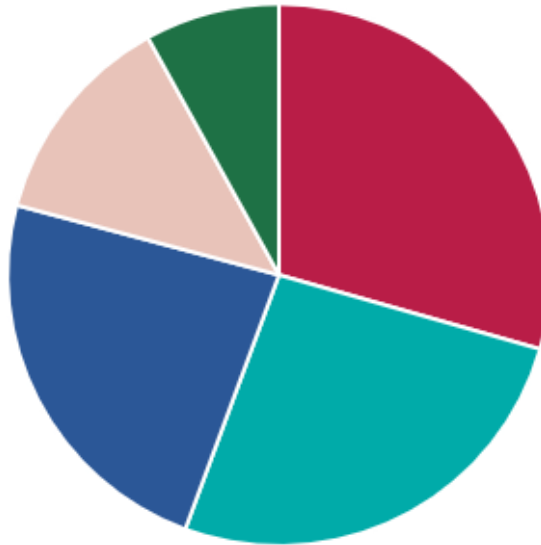
Horizontal Bars

Just change type from "bar" to "horizontalBar":

```
type: "horizontalBar",
```

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Italy France Spain USA Argentina



Example

```
new Chart("myChart", {  
  type: "pie",  
  data: {  
    labels: xValues,  
    datasets: [{  
      backgroundColor: barColors,  
      data: yValues  
    }]  
  },  
  options: {  
    title: {  
      display: true,  
      text: "World Wide Wine Production"  
    }  
  }  
});
```

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Italy France Spain USA Argentina



Just change type from "pie" to "doughnut":

```
type: "doughnut";
```

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